



Disentangling degree and tie strength heterogeneity in egocentric social networks

Sara Heydari^{1*} , Gerardo Iñiguez^{2,3,1,4}, János Kertész^{3,5} and Jari Saramäki^{1*}

*Correspondence:

sara.heydari@aalto.fi;

jari.saramaki@aalto.fi

¹ Department of Computer Science,
Aalto University School of Science,
00076 Aalto, Finland

Full list of author information is
available at the end of the article

Abstract

The structure of personal networks reflects how we organise and maintain social relationships. The distribution of tie strengths in personal networks is heterogeneous, with a few close, emotionally intense relationships and a larger number of weaker ties. Recent results indicate this feature is universal across communication channels. Within this general pattern, there is a substantial and persistent inter-individual variation that is also similarly distributed among channels. The reason for the observed universality is yet unclear—one possibility is that people's traits determine their personal network features on any channel. To address this hypothesis, we need to compare an individual's personal networks across channels, which is a non-trivial task: while we are interested in measuring the differences in tie strength heterogeneity, personal network size is also expected to vary a lot across channels. Therefore, for any measure that compares personal networks, one needs to understand the sensitivity with respect to network size. Here, we study different measures of personal network similarity and show that a recently introduced alter-preferentiality parameter and the Gini coefficient are equally suitable measures for tie strength heterogeneity, as they are fairly insensitive to differences in network size. With these measures, we show that the earlier observed individual-level persistence of personal network structure cannot be attributed to network size stability alone, but that the tie strength heterogeneity is persistent too. We also demonstrate the effectiveness of the two measures on multichannel data, where tie strength heterogeneity in personal networks is seen to moderately correlate for the same users across two communication channels (calls and text messages).

Keywords: Social network analysis; Personal networks; Egocentric networks; Tie strength heterogeneity; Social signatures; Persistence

1 Introduction

Social relationships play a fundamentally important role in our lives, both individually and collectively. On a personal level, they form a person's social capital [39], offering support, engagement, and access to resources and commodities [5]. On a societal level, they maintain cohesion and ensure the functionality of the society [9]. Social relationships vary in their levels of emotional closeness [14, 22, 36]: personal social networks typically consist of a handful of close and emotionally intense relationships (strong ties) and a larger num-

© The Author(s) 2024. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

ber of less close relationships (weak ties) (see, e.g. [12, 46]). The strong ties at the center of the network are costly to maintain and therefore limited in number. At the same time, they are essential for our and well-being [18, 19, 29, 31, 51]. The weaker ties that surround this core are more prone to network turnover [17, 44] and they are thought to provide diversity and opportunities. Their role is particularly pronounced on the network level, where they act as bridges between otherwise separate parts of the social web [10, 14, 25, 36].

The diversity of tie strengths can be quantified by studying egocentric networks derived from empirical data on social interactions [15–17, 22, 34, 44]. An egocentric network (ego-net) represents the personal network of an individual: it consists of the focal individual (the ego) and the social ties connecting the ego to its direct network neighbors (alters) [32]. Such networks can be readily constructed from communication data, where the frequency of interaction is typically used as a proxy of tie strength [37]. Several studies have provided evidence of tie strength heterogeneity in ego-nets inferred from mobile phone data [16, 28, 30, 44], with recent findings indicating strong similarities in egocentric network structure across communication channels on multiple online platforms [20].

Importantly, it has been observed in many studies that the extent of tie strength heterogeneity in ego-nets varies from person to person [8, 16, 20, 24, 28, 44]. These personal patterns of heterogeneity—social signatures—retain their characteristic shape over time even under significant network turnover, with variations that are smaller than the typical variation in the population. Furthermore, a recent study [20] has shown that the distribution of individual patterns of heterogeneity is similar across different communication channels. This is surprising because the channels serve very different purposes: while mobile-phone calls reflect offline social network structure, online conversations on websites may happen between individuals who only know each other through their usernames.

One candidate explanation for this surprising universality – of both tie strength heterogeneity and of the distribution of individual variation across channels – is that it reflects some ego-level latent variable, whose distribution in the population determines the variety of ego-net tie strengths. Here, personality traits are a plausible candidate: they are known to influence social and spatial behaviour [2], patterns of social media use [3], and mobile-phone ego-net stability and turnover [8]. However, in addition to personality traits, there are important variables such as age and gender (see, e.g., [15]). This complicates things because there are cross-correlations: e.g., observed effects of extraversion on network size appear to stem only from its correlation with age [40]. Finally, people also communicate with varying levels of intensity on different communication channels [30, 47], resulting in variations in network size that obscure differences in tie strength heterogeneity.

To better understand the behavior of tie strength heterogeneity under different circumstances, we need a method for comparing personal networks that accounts for the similarity of tie strength heterogeneity despite differences in network size. This would also help clarify whether the methods used in earlier work (e.g., [44]) truly capture the persistence of tie strength heterogeneity or simply reflect the persistence of ego-net size. To this end, we compare four measures of differences in heterogeneity between pairs of ego-nets and examine their sensitivity to ego-net size—the L^2 norm, the Jensen-Shannon divergence, and the Gini coefficient for directly comparing social signatures, and an alter-preferentiality parameter whose values are fitted with a model of ego-network evolution [20]. We demonstrate that the latter two measures are less sensitive to ego-net degree, therefore being a better option for comparing ego-nets of different sizes. We also apply these two mea-

asures to multichannel data on mobile telephone call and text message (SMS) logs, finding that the heterogeneity levels of egos are moderately correlated across these two channels. This observation supports the notion of an individual-level latent variable shaping ego-net characteristics and evolution.

2 Data and methods

2.1 Data

We use two mobile phone datasets that both include call and short message (SMS) logs. The larger dataset contains call data for more than five million users over six months and dates back to 2007. At this time, mobile phone calls and short messages were used widely, as smartphone-based alternatives for personal communication did not yet exist. We refer to the larger data as the Large Mobile Phone (LMP) dataset. We use this data set to examine how sensitive the tie-strength-based similarity measures of ego-nets are to their difference in size. The smaller dataset is known as the Copenhagen Networks Study (CNS) dataset [43] and consists of communication logs between around one thousand university students for one month during the 2012-2013 academic year. This data set is used together with the LMP set to compare heterogeneity across different communication channels. For a detailed description of these datasets, see Supplementary Information (SI) SI Section S1.1 and SI Section S1.2 (see Additional file 1).

2.2 Egocentric networks and social signatures

An egocentric network comprises an ego and the ego's alters, i.e., friends, acquaintances, and family members who have direct social ties with the ego. In weighted ego-nets, link weights represent tie strength, thus being proxies of relationship intensity. There are several approaches for defining the weights when using communication logs [16, 49]. Here, we use the communication frequency during a time window (accumulated numbers of calls or messages) as the link weights, similarly to [34, 37, 44, 50].

The concept of *social signatures* was introduced [44] for capturing individual patterns of tie strength heterogeneity. An ego's social signature is defined as the fraction of communication dedicated to each alter, ranked in decreasing order of communication intensity. For the weighted ego-net of ego e , where w_{ei} denotes the link weight of the i th most contacted alter, the vector representing the social signature is defined as

$$s_e = (w_{e1} / \sum_{i=1}^k w_{ei}, \dots, w_{ek} / \sum_{i=1}^k w_{ei}), \quad (1)$$

where k is the degree (total number of alters) of the ego e . Social signatures have been shown to be persistent over time, even under network turnover [44]. Moreover, people's call and SMS signatures have been shown to resemble each other, providing further evidence for the existence of distinct individual social signatures [16].

2.3 Measures of ego-net structural distance

People vary in the size of their social networks, how actively they use communication channels, and how they distribute their communication resources among their alters. Here, we will review four different functions for measuring the structural distances between weighted ego-nets. Our selection criterion is that each of the functions has been

used in the literature for the task. The four functions account for tie strength heterogeneity in different ways. In the following sections (3.2 and S3), we will investigate whether the observations of ego-net persistence and similarity across channels reported in [16, 20, 28, 44] hold when network size is controlled for. We will also ask how independent the observations are on the chosen distance measure.

L^2 distance The L^2 distance is a simple Euclidean distance between the social signature vectors. To measure the L^2 distance between two social signatures, we first zero-pad the shorter signature, appending zeroes to its end to ensure that both signatures are of the same length. Then, the L^2 distance between social signatures s_1 and s_2 is defined as

$$L^2(s_1, s_2) = \sqrt{\sum_{r=1}^k (f_{1r} - f_{2r})^2}, \quad (2)$$

where f_{1r} is the fraction of communication that the alter of rank r in signature s_1 receives.

Jensen-Shannon distance The elements (normalized) social signatures can also be interpreted as probabilities, so that the element i corresponds to the probability of the next communication event being targeted at the alter occupying rank i . This motivates the use of the Jensen-Shannon distance, which is an information-theoretic distance measure for comparing probability distributions. To measure the Jensen-Shannon distance between two social signatures, we again first zero-pad the shorter signature so that both of the signatures are of the same length. Then, the JSD distance d_{JSD} between social signatures s_1 and s_2 is defined as

$$d_{\text{JSD}}(s_1, s_2) = \left[H\left(\frac{1}{2}s_1 + \frac{1}{2}s_2\right) - \frac{1}{2}(H(s_1) + H(s_2)) \right]^{\frac{1}{2}}, \quad (3)$$

where $H(s_1)$ is the Shannon entropy of s_1 :

$$H(s_1) = - \sum_{r=1}^k f_{1r} \ln f_{1r}, \quad (4)$$

where k is the maximum rank in s_1 and f_{1r} the fraction of communication dedicated to alter of rank r . $f_{1r} \ln f_{1r}$ is taken as 0 for $f_{1r} = 0$.

Gini coefficient distance The Gini coefficient [13] is a measure of statistical dispersion that was originally designed to quantify inequality of wealth and income, with values ranging from 0, indicating complete equality, to 1, indicating complete inequality. In our study, we utilize the Gini coefficient to quantify the inequality in the distribution of communication among the alters of an ego similarly to [6]. In other words, the Gini coefficient compresses a social signature into a single number that represents its dispersion.

For a weighted ego-net of size k whose link weights are sorted in ascending order ($w_1 < w_2 < \dots < w_k$), the Gini coefficient G is calculated as

$$G = \frac{2 \sum_{i=1}^k i w_i}{k \sum_{i=1}^k w_i} - \frac{k+1}{k}. \quad (5)$$

We define the metric distance between a pair of Gini coefficients G_1 and G_2 as

$$\Delta G(G_1, G_2) = |G_1 - G_2|. \quad (6)$$

Alter-preferentiality distance The alter-preferentiality parameter β was introduced as part of an ego-net growth model [20]. In the model, β is a tuning parameter that determines the tendency of an ego to preferentially contact alters with more previous contacts, resulting in a “rich-gets-richer” phenomenon. The values of β range from zero to infinity, with the rich-get-richer phenomenon dominating for $\beta > 1$, resulting in heterogeneous tie strengths. For $\beta < 1$, contacts take place randomly, resulting in homogeneous tie strengths. To estimate the alter preferentially parameter β associated with a weighted ego-net, we utilize a maximum likelihood estimation method and a goodness-of-fit (GOF) test, as described in Ref. [20]. Our GOF test uses the Kolmogorov-Smirnov statistic [38] with a p-value threshold of 0.1. As all fitted values may not pass the test, the procedure may not assign a value to an ego-net. For our empirical data, the percentages of egos with valid fits are presented in Table S1. Furthermore, after fitting the values of β , we filter out outliers by excluding the ego-nets with the largest five percent of values of β , except for Fig. 5 due to the small size of the CNS dataset.

Unlike the Gini coefficient which has a bounded range, β has no upper bound. Therefore, we propose a normalized distance function for measuring the distance between two ego-nets with alter-preferentiality parameters β_1 and β_2 as

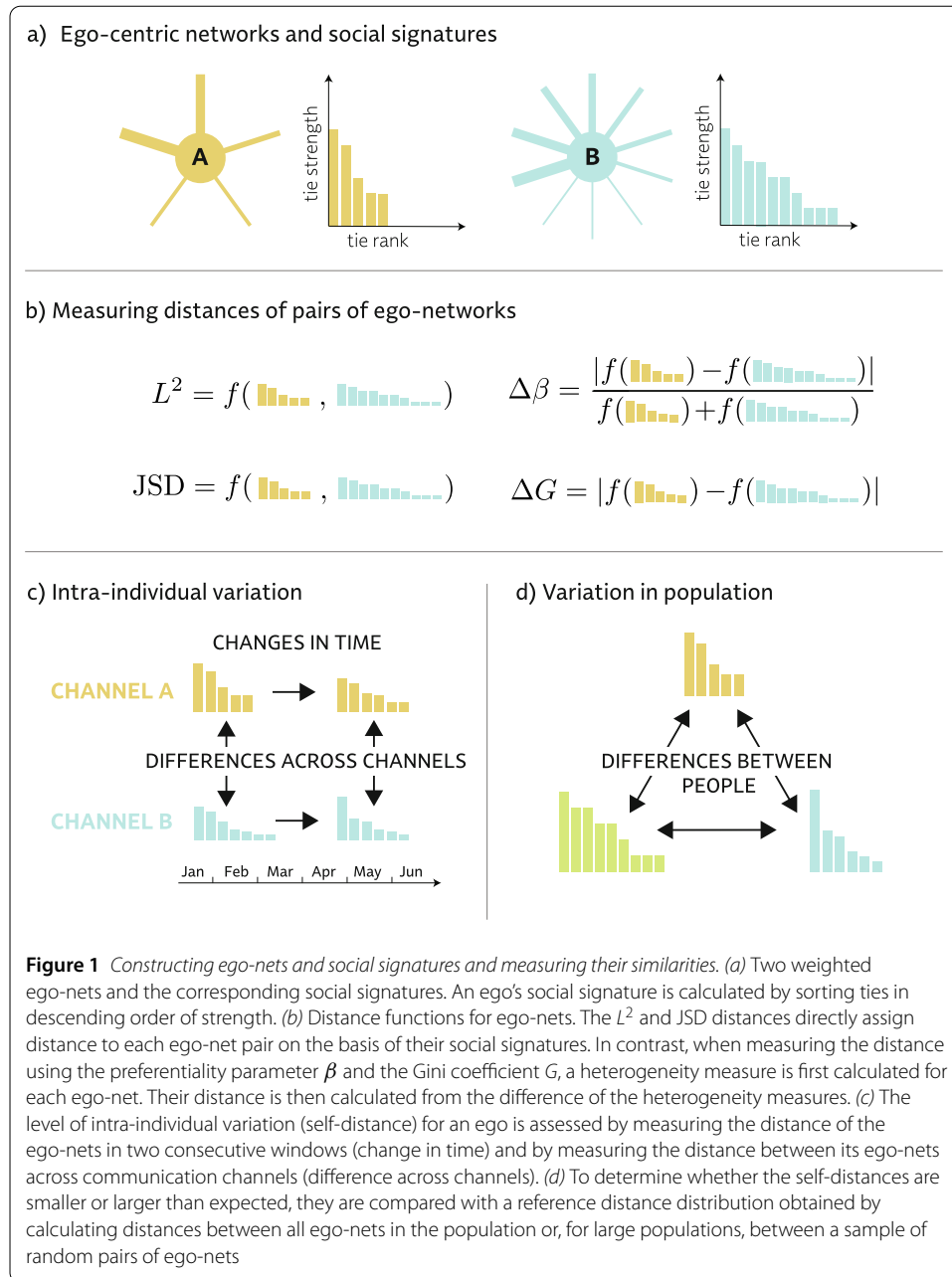
$$\Delta\beta(\beta_1, \beta_2) = \frac{|\beta_1 - \beta_2|}{\beta_1 + \beta_2}. \quad (7)$$

Similarly to Eq. (6), this normalized distance is bounded between zero and one.

2.4 Persistence of social signatures revisited

Egocentric networks are dynamic: people initiate new social relationships and lose contact with old acquaintances over time [41, 44]. Over shorter time scales, social ties display bursty patterns of activity, with bursts of frequent contact that are separated by longer periods of silence [4, 7, 21, 35]. Despite the constant changes and turnover in network membership, several studies have reported structural persistence of ego-nets [16, 28, 44]. There are two main characteristics of weighted ego-nets: their size (the number of alters) and how the limited communication resources are distributed among the alters (the shape of the social signature). As these two aspects are somewhat entangled, it is not entirely clear whether the ego-net persistence reported in the literature is merely due to stability in ego-net size or whether individual patterns of tie strength heterogeneity are also persistent.

The approach taken in earlier literature is to compare temporal changes in individual egonets with the amount of variation in the population to determine whether the changes are small or large [8, 16, 20, 24, 28, 44]), see Fig. 1 c) and d). In this study, we modify this approach to disentangle the effects of degree stability and the persistence of tie strength heterogeneity. To measure temporal changes, we divide each ego’s timeline into two consecutive windows. Here, instead of having consecutive windows with equal duration, the window sizes are determined by the number of communication events so that there are equal numbers of communication events in both windows. Then, in line with previous



studies, we construct the weighted ego-net associated with each window and measure the structural difference between the two ego-nets, $d(ew_1, ew_2)$, using the different distance functions defined in Sect. 2.3.

To investigate the effects of changes in ego-net size, we compare the values of the distance functions to two different reference distributions. The first one is a general population reference with no constraint (the approach taken in the earlier studies [8, 16, 20, 24, 28, 44]). The second one is a degree-stratified reference consisting exclusively of ego pairs that have a difference in ego-net size equal to the size difference of the focal ego for the two windows. In the first approach, the reference distribution comprises the ego-net distances between randomly selected egos x in the first period and randomly selected egos y

in the second period, without paying attention to their degrees (with $x \neq y$). In the second method, we form the reference distribution of a subset of these instances, the pairs (x, y) that satisfy the condition $|k_{e1} - k_{e2}| = |k_{x1} - k_{y2}|$ where k_{e1} and k_{e2} are the degrees of ego e in the different time windows and k_{x1} and k_{y2} are the degrees of egos x and y in their respective time windows.

To quantify the persistence of the network of ego e , we calculate the z -score, a signed dimensionless quantity that measures the deviation of $d(ew_1, ew_2)$ from the mean value of the reference distribution of choice, ref , in units of the standard deviation of ref :

$$z(d(ew_1, ew_2), \text{ref}) = \frac{d(ew_1, ew_2) - \text{avg}(\text{ref})}{\text{std}(\text{ref})}. \quad (8)$$

A negative value means that the temporal change for the ego is smaller than the average distance between a random ego in the first window and another random ego in the second window. This non-randomness is taken as indicative of ego-net persistence, with larger absolute values of the z -score indicating stronger persistence.

3 Results

To investigate whether weighted ego-nets display persistence even after their sizes are controlled for, we use the LMP dataset (see S1.1) which includes communication logs of over four million users for our investigations. This size facilitates computing degree-stratified reference distributions, as pairs of egos with a chosen degree difference can almost always be found, and guarantees the significance of findings, as the analysis is based on a very large number of data points.

This section is organized as follows: first, in Sect. 3.1, we establish our motivation for probing the concept of ego-net stability by showing that the distance functions introduced in Sect. 2.3 are all to some extent sensitive to the degree differences of the ego-net pairs. We also show that compared with the variation in the population, temporal variation in the size of individual ego-nets is small. Therefore, it is important to verify that earlier results are not only due to this degree persistence.

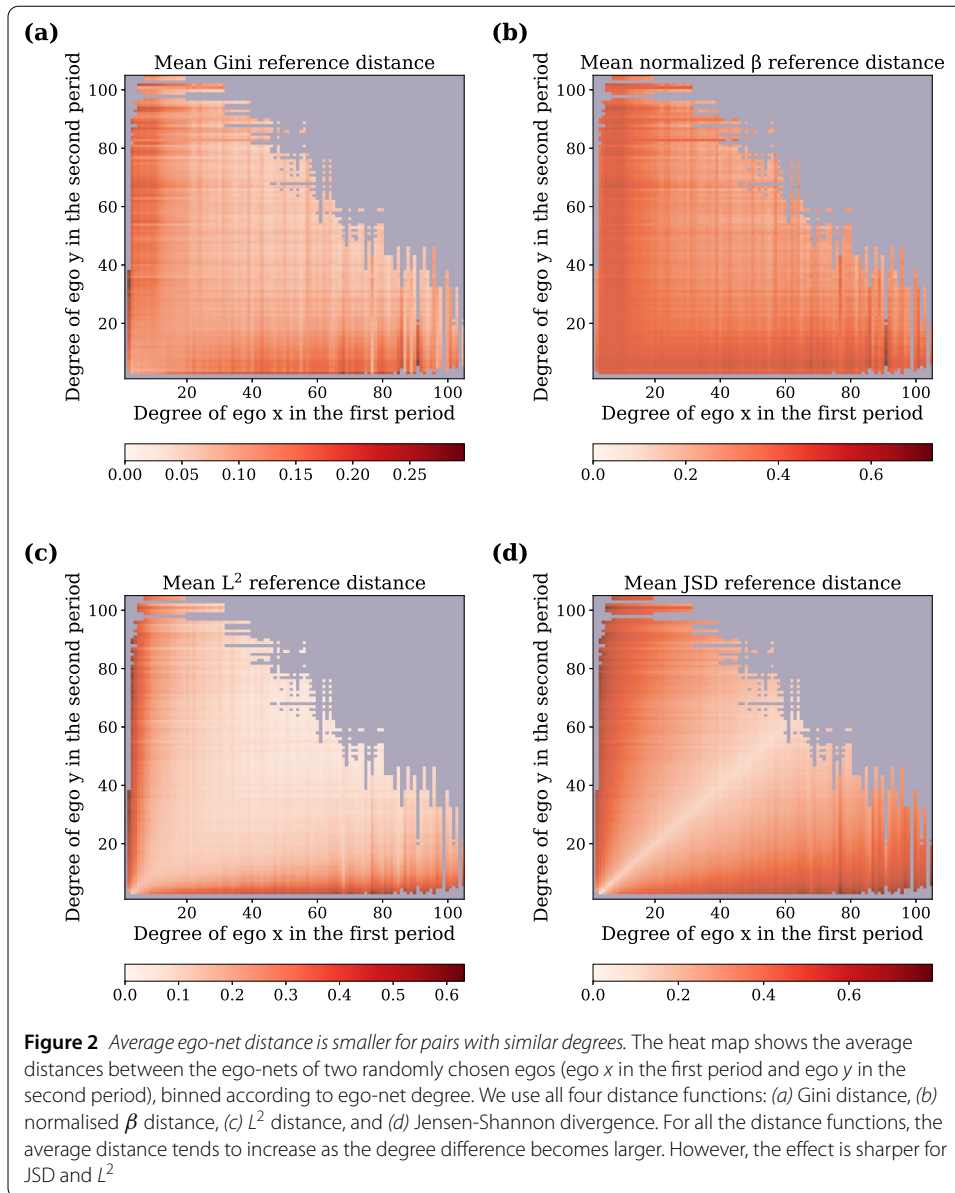
In Sect. 3.2 we show that this is not the case: persistence of social signatures reported in the literature holds even for the degree-controlled reference, indicating that ego-nets have persistent patterns of tie strength heterogeneity.

We also observe that among the four structural distance functions used in the literature (see Sect. 2.3), the distance functions based on the Gini coefficient and the alter preferentially β are less sensitive to differences in ego-net size. Thus, they are more appropriate measures for comparing ego-nets of different sizes if we wish to focus on tie strength heterogeneity.

Finally, using these two degree-insensitive measures, we investigate the similarity of ego-centric networks across the call and SMS channels. For this purpose, we use the CNS and LMP datasets (see Sects. S1.2 and S1.1 for data description). This comparison suggests that the heterogeneity levels of ego-nets are moderately correlated across call and SMS channels, in line with the results in Ref. [16] where degrees were not controlled for.

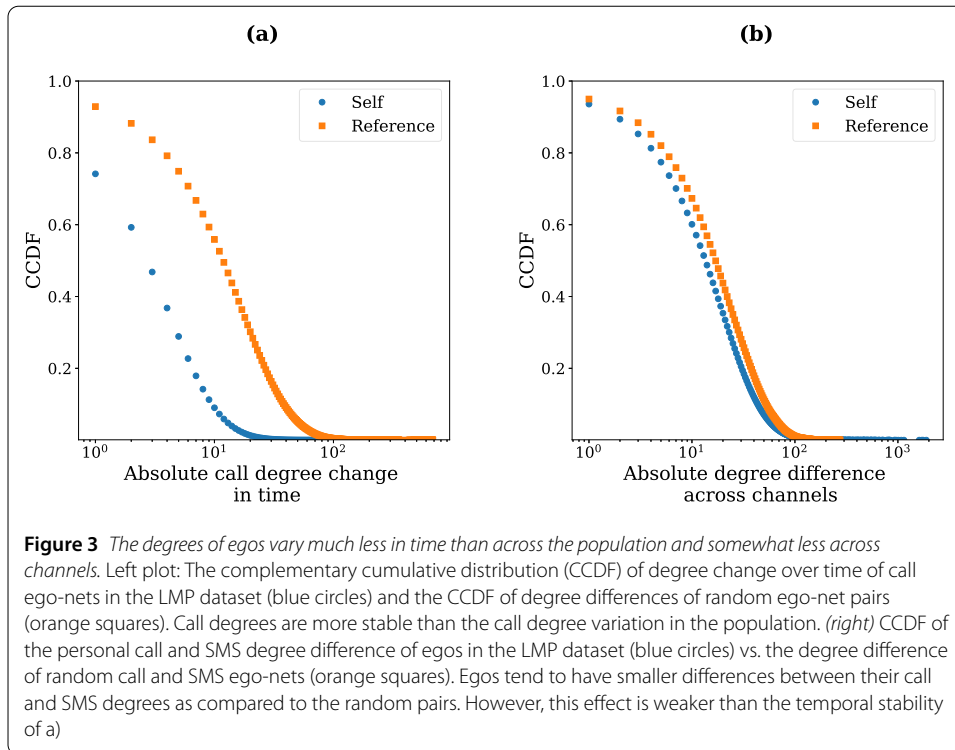
3.1 Disentangling the persistence of ego-net size and tie strength heterogeneity

We start by empirically investigating how the distance measures (introduced in Sect. 2.3) depend on the degree difference of the ego-net pairs. To this end, we take a random sample



of approximately 7000 egos from the LMP dataset and form all possible ordered ego pairs ($\sim 7000 \times 6999$ pairs). For each ego pair (e_1, e_2) , we construct an ego-net for e_1 based on the first half of its call log and an ego-net for e_2 using the second half of e_2 's call log. Then we calculate the distance between these pairs of ego-nets using each of the four distance functions defined in Sect. 2.3 (note that these are the same ego-nets that we will use to form the reference distribution in Sect. 3.2).

To investigate how the measured distance depends on the ego-net size (degree), we group the ego-net pairs based on their degrees (k_1, k_2) and calculate the average distance for each group and each distance function. Figure 2 shows a heat map of the average distances for each group as a function of the degree of the ego-nets for each distance function. To ensure significance of findings, bins with less than 200 pairs are not shown. The plots indicate that for all distance functions, the measured distance tends to be smaller when the ego-nets have similar degrees (lighter colours closer to the diagonal and darker colours



further from it). However, the effect is clearly much stronger for the two signature-based distance functions (JSD and L^2).

As the structural distance of ego-nets and degree difference appear correlated, the next natural step is to study the variation of degree. Figure 3 shows the complementary cumulative distributions of both the amount of individual degree change of ego-nets in time and the degree differences of each individual's ego-nets across channels. As a reference distribution, we use the degree differences of random ego-net pairs (variation in the population). We observe that compared to the reference, ego-nets tend to have more stable degrees in time. Across-channel differences are on average also below the population-level reference, though the difference is far less pronounced.

Together with the degree sensitivity of the distance functions (Fig. 2), the result of Fig. 3 prompts us to revisit the notion of ego-net persistence and quantify how it is affected by degree differences: is structural heterogeneity also persistent, or are the earlier results on ego-net persistence only due to the persistence of ego-net size?

3.2 Impact of degree changes on the persistence of ego-nets

Next, we investigate the impact of degree changes on the persistence of ego-nets when using different functions to measure their distance (see Sect. 2.3 for a description of different distance functions and Sect. 2.4 on how to quantify persistence). For this analysis, we again use the call log data from the LMP dataset. The extensive size of the dataset enables us to draw a sufficiently large sample for establishing a representative reference distribution of distances in the population. To quantify persistence, we compare the self-distances that represent the structural changes of each ego-net from one period to another against the reference distribution and calculate the corresponding z-scores [see Eq. (8)].

When calculating the self-distances, we limit ourselves to the egos to whom we can assign values for both the preferentiality parameter β and the Gini coefficient, i.e. those ego-nets that pass the goodness-of-fit test for β and have degrees larger than one so it is possible to calculate their Gini coefficient. This leaves us with around 3 million ego-nets.

To determine whether the self-distances are smaller or larger than expected on the basis of population-level variation, as explained in Sect. 2.4, we use two population references. The first is a general, degree-agnostic reference and the second is degree-stratified, based on groups of ego-net pairs with similar values of $|\Delta k|$. The sample for construction the degree-agnostic reference is the same as in 3.1 ($\sim 49\text{M}$ pairs). For the $|\Delta k|$ -specific references, we use a specific subset of the sample determined by the degree change of the ego whose ego-net persistence we are assessing (See 2.4).

With the self-distances and the reference distributions, we can measure the persistence score of each ego using Eq. (8). Figure 4 shows the distribution of persistence scores of call ego-nets among the population (around three million egos) in the LMP dataset. The persistence values are calculated using all four distance functions introduced in Sect. 2.3, against a single degree-agnostic reference [panel (a)] as well as $|\Delta k|$ -specific references [panel (b)]. We observe that for all these cases, most egos have negative persistence scores, indicating the stability of the ego-nets. Therefore, one can conclude that the earlier results in the literature also hold for the persistence of tie strength heterogeneity.

To examine in detail the impact of degree change on the persistence of the ego-nets, we take a closer look at the persistence scores by grouping the egos based on the absolute value of their degree change, denoted as $|\Delta(k)|$, and calculate the average persistence for each group separately (see Fig. 4). We observe that the commonly used JSD distance depends rather strongly on the degree difference, with the curves for the degree-agnostic and degree-stratified reference deviating to a large extent for large degree differences. On the other hand, the β and Gini distances behave much more smoothly, with only small differences between the degree-agnostic and degree-stratified persistence scores. This result indicates that the alter preferentiality β and the Gini inequality measure G are both fairly insensitive to degree difference. Therefore, these two measures can be considered more suitable for comparing ego-nets with varying sizes.

3.3 Application: similarity of ego-nets across call and SMS channels using the Gini coefficient and the alter preferentiality β

We demonstrate the use of the preferentiality parameter β and the Gini coefficient by comparing ego-nets across two communication channels (calls and SMS). This question is of interest for inference of the underlying, “real-world” social network: both channels are used for everyday communication (unlike, e.g., specialized online sites), and the ego-nets constructed from the respective logs can be expected to reflect the underlying network, at least when it comes strong ties with repeated communication events. Naturally, channel preferences differ, and communication on one single channel only represents an incomplete reflection of the underlying social network. However, if the underlying social network is the latent driver of communication on both channels, an individual’s call and text ego-nets should resemble one another. This was observed earlier in [16], however using the degree-sensitive JSD distance between social signatures. Here, our aim is to see whether the results hold for degree-insensitive distance measures, indicating similarity of tie strength heterogeneity.

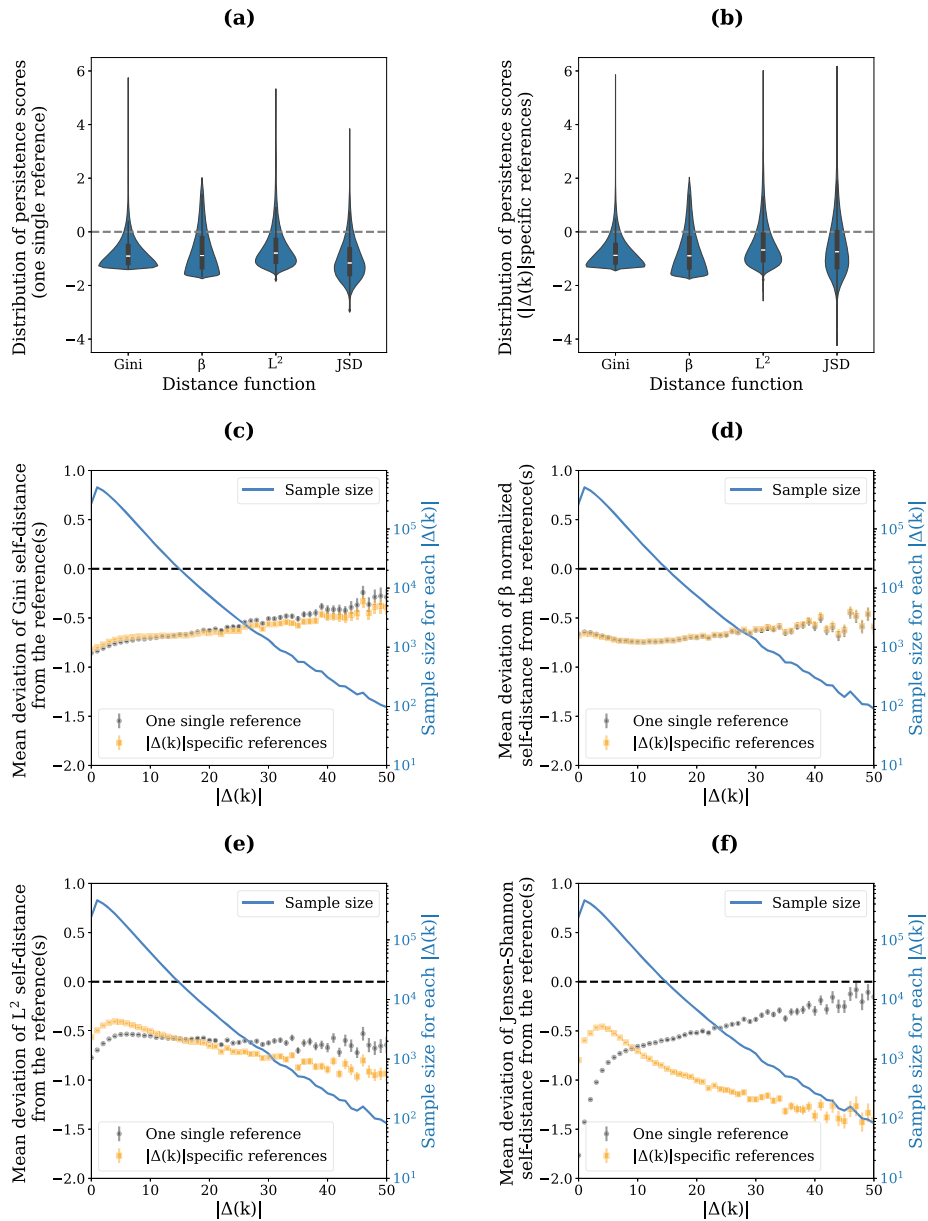
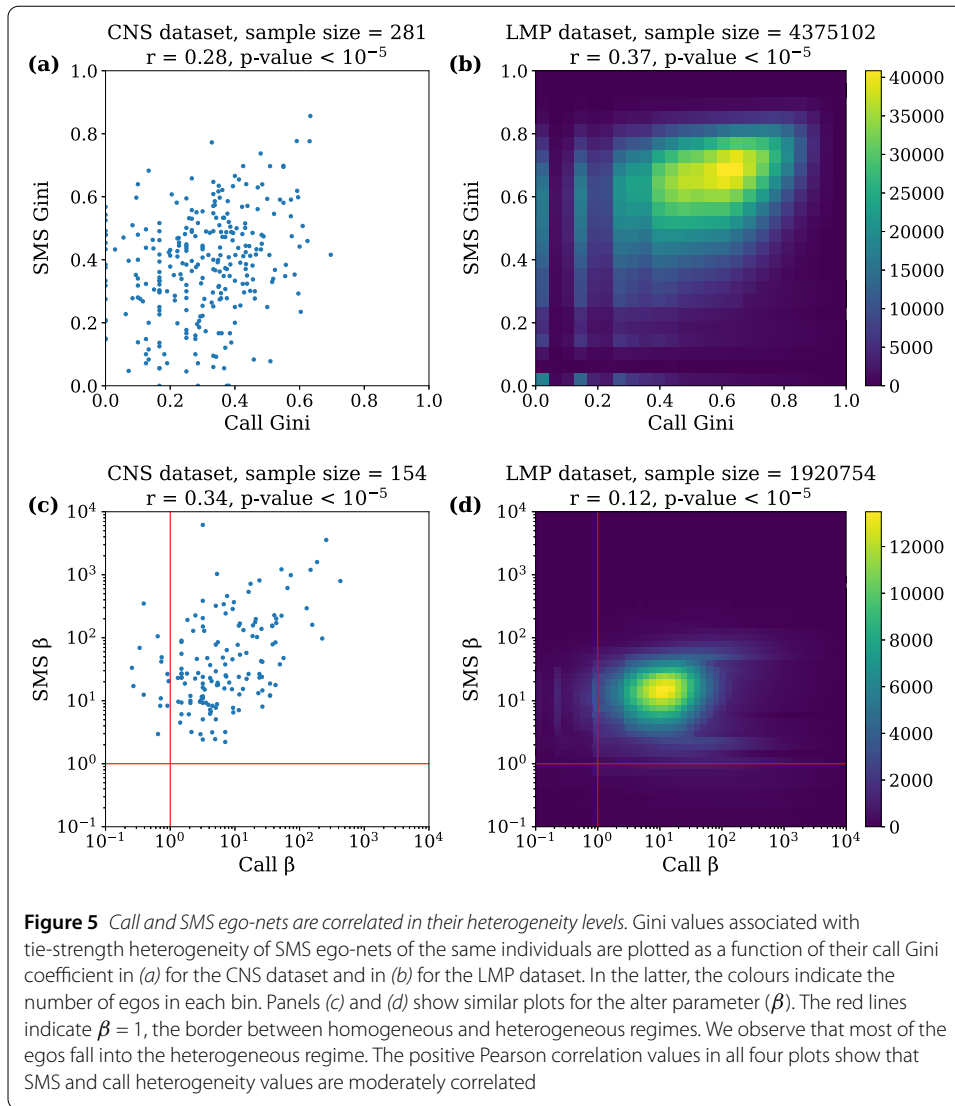


Figure 4 Structural persistence of call ego-nets in the LMP dataset using different functions for measuring structural distance. (a) Distribution of persistence scores in units of standard deviations in the LMP dataset when measured against one single reference and in (b) when measured against $|\Delta(k)|$ -specific references. In both cases, most of the scores are negative, which indicates that call-egonets show structural persistence. The average persistence scores as a function of $|\Delta(k)|$ (absolute degree change) are shown for (c) the Gini distance function; (d) the normalised β distance; (e) the L^2 distance; and for (f) the JSD distance. These scores measure the deviation of self-distances from reference distances and they are shown in grey use one single reference distribution (regardless of $|\Delta(k)|$) and in orange for $|\Delta(k)|$ -specific reference distributions. See Eq. (8) for the definition of the persistence score. All four heterogeneity measures have negative average values across the whole degree range, indicating structural persistence. Moreover, the curves for the Gini coefficient and β do not change much when switching from a single reference to $|\Delta(k)|$ -specific references. This suggests that Gini and β are better measures for comparing ego-nets of different sizes. The blue curves in plots (a), (b), (c), and (d) show the number of egos with a particular absolute degree change ($|\Delta(k)|$)



We start with the LMP dataset and assess the similarity of ego-nets across channels with an approach that controls for different levels of degree difference (similar to our approach in analyzing the persistence of ego-nets in Sect. 3.2). We conclude that call and SMS ego-nets resemble each other, but the effect is smaller than the observed persistence of ego-nets in time (see Figure S4 and Section S3).

Next, as the Gini coefficient and the alter preferentiality β do not require the use of reference distributions, we directly correlate the Gini and β values for call and SMS ego-nets for both of our datasets (LMP and CNS), as shown in Fig. 5. We observe moderate correlations across the communication modalities, indicating that there are clear similarities between an individual's tie strength heterogeneity across the channels. This qualitatively confirms the findings of [16].

Taken together, the above results support the hypothesis of a latent driver or multiple latent drivers of ego-net structure. For calls and text messages, the underlying “real-world” social network is an obvious candidate, with strong ties giving rise to frequent communication on both channels. However, as discussed in [16], the situation is not this simple: the

alter composition of call and text ego-nets can vary considerably, leaving room for other latent variables such as personality traits: if an individual's call and text ego nets contain different alters and still display similar levels of tie strength heterogeneity, it is reasonable to expect that there is a latent variable behind this.

3.4 Discussion

Egocentric networks are the building blocks of societal-level networks. They comprise a wide variety of relationships ranging from romantic partners and family to friends, colleagues, and acquaintances. This diversity of relationship types is reflected in the structure of ego-nets: not all social ties are equally strong. Rather, there is a heterogeneous structure with a limited number of strong ties at the core of the network, surrounded by larger number of weaker ties. This pattern represents a tradeoff: while strong ties are crucial to our health and well-being [18, 19, 31], they demand significant time and emotional investment [11, 12]. Weaker ties require less effort, but are nevertheless important, e.g., for access to new information and generally, for societal cohesion [14]. This characteristic pattern has been observed in social networks inferred from various communication datasets [16, 20, 34, 44]. Within this broad pattern, there is, however, individual-level variation, as people express different levels of heterogeneity in their personal networks. Besides tie strength heterogeneity, ego-nets vary in size, i.e., network degrees (see, e.g., [37]). The size of an empirically observed ego-net may reflect the size of the ego's underlying social network, but it may also be affected by issues such as sampling, the length of the observation window, and the ego's preferential channel choices [27, 47]).

Even though the literature mostly addresses tie strengths from the perspective of social relationships in general ("offline relationships"), ego-nets display a similar pattern of tie strength heterogeneity on many communication channels and online platforms, including ones that cannot be expected to reflect offline relationships [20]. It is not only the broad pattern but the distribution of individual variation within the pattern that appears universal. This may be a population-level consequence of people having different channel preferences and behaving differently on different channels so that the mixture of these different behaviours is behind the distribution of heterogeneity. Alternatively, the distribution may reflect the distribution of some latent individual-level variables in the population, such as personality traits, which determine people's ego-net structure on any channel. However, somewhat surprisingly, attempts to establish a direct link between the Big Five personality traits and ego-net structure have not been successful, with the exception of network turnover [8, 40].

Both ego-net size and tie strengths are encapsulated in so-called social signatures [44] that quantify the share of communication that alters receive as a function of their rank in an ego-net. These signatures are rather persistent for an individual, with variations smaller than the typical population-level variation, even when the composition of alters in the network changes [44]. An individual's signatures derived from call and SMS logs also resemble each other [16]. However, until now, it has not been established whether this is only due to persistence of ego-net size, or whether patterns of tie strength heterogeneity are equally persistent. In this paper, we carried out an analysis accounting for the sensitivity on degree differences of distance functions used for comparing ego-nets. We showed that the observed persistence of social signatures and their individual-level similarity across the call and SMS channels are not merely a side effect of the stability of the network size.

Our comparison of the distance functions also revealed that two measures are far less sensitive to degree differences: the alter preferentiality β and the Gini coefficient. These two are thus better options for assessing the persistence of tie strength heterogeneity when network sizes are also heterogeneous and for understanding whether the same individuals have similar ego-net shapes on different communication channels, where network sizes may generally vary quite extensively. These measures are highly correlated and theoretically related (see S2.1), and therefore, either of them can be used. The Gini coefficient is a well-known statistical indicator and is also straightforward to compute. On the other hand, the alter-preferentiality parameter β has more explanatory power since it is associated with an ego-network growth model. However, the required fitting procedure is rather cumbersome, though it has the benefit of automatically filtering out ego-nets that are of too low degree or entirely homogeneous (see Fig. S3).

Another advantage is that both coefficients attribute a heterogeneity value to each ego-net, facilitating a more straightforward comparison than when using signature-based distance functions (e.g. JSD and L^2). This is particularly relevant for comparing ego-nets across different communication modalities. We demonstrated this by applying the Gini and β measures to investigate if the call and SMS ego-nets of an individual are correlated in terms of tie strength heterogeneity. We used two datasets that contain both calls and SMS messages and observed that the tie strength heterogeneity measures show a moderate correlation for call and SMS ego-nets.

Taken together, our results for call and SMS ego-nets indicate a moderate level of similarity in terms of tie strength heterogeneity, confirming the results of [16]. Obviously, both call and SMS ego-nets can be expected to reflect an individual's offline social relationships. However, as shown in [16], these networks may only partially overlap in terms of alters and shared alters may be ranked very differently. Furthermore, the two channels may serve somewhat different purposes, as the typical temporal patterns of calling and texting have been observed to significantly differ [1]. Therefore, their similarity hints at the existence of other latent variables, such as personality traits, that drive ego-net structure on multiple platforms.

More generally, our observations, along with findings from earlier literature [8, 20, 24, 28, 44, 47] indicate that ego-nets typically exhibit similar, persistent, heterogeneous patterns of tie strength across multiple platforms. There are many open questions: does the moderate similarity of call and SMS ego-nets generalize to other platforms, or is it primarily due to both reflecting the shape of offline networks? In other words, do an individual's ego-nets resemble one another because of some latent variable(s), whatever the platform, or are there different mechanics at play: e.g., on an online platform where discussion participants only see each others' nicknames, tie strength heterogeneity might arise simply because of self-reinforcing processes with memory, such as engaging in repeated discussions with the same people for convenience. To answer these questions, further research and data collection efforts are required: besides call and text logs, there are no multi-platform data sets for the same users. This highlights the need for experiments where the same participants are followed across multiple platforms and surveyed, e.g., for the Big Five personality traits to address the latent-variable hypothesis.

Abbreviations

Ego-net, egocentric network; SMS, a type of text message transmitted via Short Messaging Service; LMP, Large Mobile Phone data set; CNS, Copenhagen Networks Study; SI, Supplementary Information; JSD, Jensen-Shannon Divergence; GOF, goodness-of-fit; CCDF, Complementary cumulative distribution function.

Supplementary information

Supplementary information accompanies this paper at <https://doi.org/10.1140/epjds/s13688-024-00513-x>.

Additional file 1. (PDF 453 kB)

Acknowledgements

We acknowledge the computational resources provided by the Aalto Science-IT project.

Author contributions

SH, GI, JK, and JS conceived, designed, and developed the study. SH analysed the empirical data. GI derived the analytical relations in Sect. S2.2. SH, GI, JK, and JS wrote the paper. All authors read and approved the final manuscript.

Funding

The study is funded by the Strategic Research Council (SRC) established within the Research Council of Finland (decision numbers: 364386 and 364371). G.I. and J.K. acknowledge support from AFOSR (Grant No. FA8655-20-1-7020), project EU H2020 Humane AI-net (Grant No. 952026), and CHIST-ERA project SAI (Grant No. FWF I 5205-N). J.K. acknowledges support from European Union's Horizon 2020 research and innovation programme under grant agreement ERC No 810115 - DYNASNET.

Availability of data and materials

The CNS data analysed in this paper is publicly available via *figshare* [42, 43]. While the LMP data is not publicly available and shared with us via a non-disclosure agreement, it has been extensively studied in the literature (see, for example, [16, 21, 23, 26, 36, 37, 48]).

Declarations

Competing interests

The authors declare that they have no competing interests.

Author details

¹Department of Computer Science, Aalto University School of Science, 00076 Aalto, Finland. ²Faculty of Information Technology and Communication Sciences, Tampere University, 33720 Tampere, Finland. ³Department of Network and Data Science, Central European University, 1100 Vienna, Austria. ⁴Centro de Ciencias de la Complejidad, Universidad Nacional Autónoma de México, 04510 Ciudad de México, Mexico. ⁵Complexity Science Hub, 1080 Vienna, Austria.

Received: 29 March 2024 Accepted: 22 November 2024 Published online: 18 December 2024

References

1. Aledavood T, López E, Roberts SG, Reed-Tsochas F, Moro E, Dunbar RI, Saramäki J (2016) Channel-specific daily patterns in mobile phone communication. In: Proceedings of ECCS 2014: European conference on complex systems. Springer, Berlin, pp 209–218
2. Alessandretti L, Lehmann S, Baronchelli A (2018) Understanding the interplay between social and spatial behaviour. *EPJ Data Sci* 7:36
3. Azucar D, Marengo D, Settanni M (2018) Predicting the big 5 personality traits from digital footprints on social media: a meta-analysis. *Pers Individ Differ* 124:150–159
4. Barabasi A-L (2005) The origin of bursts and heavy tails in human dynamics. *Nature* 435(7039):207–211
5. Berkman L, Glass T, Brissette I, Seeman T (2000) From social integration to health: Durkheim in the new millennium. *Soc Sci Med* 51:843–857
6. Bhattacharya K, Ghosh A, Monsivais D, Dunbar RI, Kaski K (2016) Sex differences in social focus across the life cycle in humans. *R Soc Open Sci* 3(4):160097
7. Candia J, González MC, Wang P, Schoenharl T, Madey G, Barabási A-L (2008) Uncovering individual and collective human dynamics from mobile phone records. *J Phys A, Math Theor* 41(22):224015
8. Centellegher S, López E, Saramäki J, Lepri B (2017) Personality traits and ego-network dynamics. *PLoS ONE* 12(3):e0173110
9. Coleman JS (1990) Foundations of social theory. Belknap Press of Harvard University Press, Cambridge
10. Csermely P (2006) Weak links: stabilizers of complex systems from proteins to social networks. Springer, Berlin
11. Dunbar R (2009) The social brain hypothesis and its implications for social evolution. *Ann Hum Biol* 36:562–572
12. Dunbar R (2018) The anatomy of friendship. *Trends Cogn Sci* 22:32–51
13. Gini C (1921) Measurement of inequality of incomes. *Econ J* 31(121):124–125
14. Granovetter MS (1973) The strength of weak ties. *Am J Sociol* 78(6):1360–1380
15. Growiec K (2023) Social network dynamics in the context of age: an empirical investigation. *Soc Indic Res* 170(2):675–711
16. Heydari S, Roberts SG, Dunbar RI, Saramäki J (2018) Multichannel social signatures and persistent features of ego networks. *Appl Netw Sci* 3:8

17. Hidd VV, López E, Centellegher S, Roberts SGB, Lepri B, Dunbar RIM (2023) The stability of transient relationships. *Sci Rep* 13(1):6120
18. Holt-Lunstad J, Smith TB, Layton JB (2010) Social relationships and mortality risk: a meta-analytic review. *PLoS Med* 7(7):e1000316
19. House JS, Landis KR, Umberson D (1988) Social relationships and health. *Science* 241(4865):540–545
20. Iñiguez G, Heydari S, Kertész J, Saramäki J (2023) Universal patterns in egocentric communication networks. *Nat Commun* 14:5217
21. Karsai M, Kivela M, Pan RK, Kaski K, Kertész J, Barabási A-L, Saramäki J (2011) Small but slow world: how network topology and burstiness slow down spreading. *Phys Rev E* 83(2):025102
22. Kempnich M, Wölfer R, Hewstone M, Dunbar R (2024) How the size and structure of egocentric networks change during a life transition. *Adv Life Course Res* 61:100632
23. Kivela M, Pan RK, Kaski K, Kertész J, Saramäki J, Karsai M (2012) Multiscale analysis of spreading in a large communication network. *J Stat Mech* 2012(3):P03005
24. Koltsova OY, Mararitsa LV, Terpilovskii MA, Sinyavskaya YE (2021) Social signature in an online environment: stability and cognitive limits. *Comput Hum Behav* 122:106856
25. Korte C, Milgram S (1970) Acquaintance linking between white and negro populations: application of the small world problem. *J Pers Soc Psychol* 15(2):101–108
26. Kovanen L, Kaski K, Kertész J, Saramäki J (2013) Temporal motifs reveal homophily, gender-specific patterns, and group talk in call sequences. *Proc Natl Acad Sci* 110(45):18070–18075
27. Krings G, Karsai M, Bernhardsson S, Blondel VD, Saramäki J (2012) Effects of time window size and placement on the structure of an aggregated communication network. *EPJ Data Sci* 1(1):4
28. Li Y, Bond RM (2022) Evidence of the persistence and consistency of social signatures. *Appl Netw Sci* 7(1):10
29. Litwin H, Stoeckel KJ (2014) Confidant network types and well-being among older Europeans. *Gerontologist* 54:762–772
30. Liu B, Xu X-K, Lü L (2024) Uncovering patterns of multichannel mobile communications. *Phys A, Stat Mech Appl* 636:129546
31. Liu H, Waite L (2014) Bad marriage, broken heart? Age and gender differences in the link between marital quality and cardiovascular risks among older adults. *J Health Soc Behav* 55:403–423
32. Marsden P (1990) Network data and measurement. *Annu Rev Sociol* 16:435–463
33. McDonald JB, Jensen BC (1979) An analysis of some properties of alternative measures of income inequality based on the gamma distribution function. *J Am Stat Assoc* 74(368):856–860
34. Miritello G, Moro E, Lara R, Martínez-López R, Belchamber J, Roberts SG, Dunbar RIM (2013) Time as a limited resource: communication strategy in mobile phone networks. *Soc Netw* 35(1):89–95
35. Oliveira JG, Barabási A-L (2005) Human dynamics: Darwin and Einstein correspondence patterns. *Nature* 437(7063):1251–1251
36. Onnela J-P, Saramäki J, Hyvönen J, Szabó G, De Menezes MA, Kaski K, Barabási A-L, Kertész J (2007) Analysis of a large-scale weighted network of one-to-one human communication. *New J Phys* 9(6):179
37. Onnela J-P, Saramäki J, Hyvönen J, Szabó G, Lazer D, Kaski K, Kertész J, Barabási A-L (2007) Structure and tie strengths in mobile communication networks. *Proc Natl Acad Sci* 104(18):7332–7336
38. Press WH, Teukolsky SA, Vetterling WT, Flannery BP (1988) Numerical recipes in C. Cambridge University Press, Cambridge
39. Putnam RD (2000) Bowling alone: the collapse and revival of the American society. Simon and Schuster, New York
40. Roberts S, Wilson R, Fedurek P, Dunbar R (2008) Individual differences and personal social network size and structure. *Pers Individ Differ* 44:954–964
41. Roy C, Bhattacharya K, Dunbar RI, Kaski K (2022) Turnover in close friendships. *Sci Rep* 12(1):11018
42. Sapiezynski P, Stopczynski A, Lassen DD, Jørgensen SL (2019) The Copenhagen Networks Study interaction data. figshare. <https://doi.org/10.6084/m9.figshare.7267433>
43. Sapiezynski P, Stopczynski A, Lassen DD, Lehmann S (2019) Interaction data from the Copenhagen networks study. *Sci Data* 6(1):1–10
44. Saramäki J, Leicht EA, López E, Roberts SG, Reed-Tsochas F, Dunbar RI (2014) Persistence of social signatures in human communication. *Proc Natl Acad Sci* 111(3):942–947
45. Stopczynski A, Sekara V, Sapiezynski P, Cuttone A, Madsen MM, Larsen JE, Lehmann S (2014) Measuring large-scale social networks with high resolution. *PLoS ONE* 9(4):e95978
46. Sutcliffe A, Dunbar R, Binder J, Arrow H (2012) Relationships and the social brain: integrating psychological and evolutionary perspectives. *Br J Psychol* 103:149–168
47. Török J, Murase Y, Jo H-H, Kertész J, Kaski K (2016) What big data tells: sampling the social network by communication channels. *Phys Rev E* 94:052319
48. Unicom S, Iñiguez G, Karsai M (2018) Threshold driven contagion on weighted networks. *Sci Rep* 8(1):1–10
49. Ureña-Carrion J, Saramäki J, Kivela M (2020) Estimating tie strength in social networks using temporal communication data. *EPJ Data Sci* 9(1):37
50. Wang D, Pedreschi D, Song C, Giannotti F, Barabasi A-L (2011) Human mobility, social ties, and link prediction. In: Proceedings of the 17th ACM SIGKDD international conference on knowledge discovery and data mining. ACM, New York, pp 1100–1108
51. Wittig RM, Crockford C, Lehmann J, Whitten PL, Seyfarth RM, Cheney DL (2008) Focused grooming networks and stress alleviation in wild female baboons. *Horm Behav* 54(1):170–177

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.